



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 2

Kirchhoff's Laws, Nodal & Mesh Analysis

Dr. Inmaculada (Inma) Tomeo-Reyes

Senior Lecturer

School of Electrical Engineering and Telecommunications, UNSW

Topic 2 Content

This lecture covers:

- Nodes, Branches and Loops/Meshes
- Kirchhoff's Laws
- Series and Parallel Connection of Circuit Elements
- Voltage and Current Division
- Nodal Analysis
- Mesh Analysis

**Corresponds to Chapters 2
and 3 of your textbook**

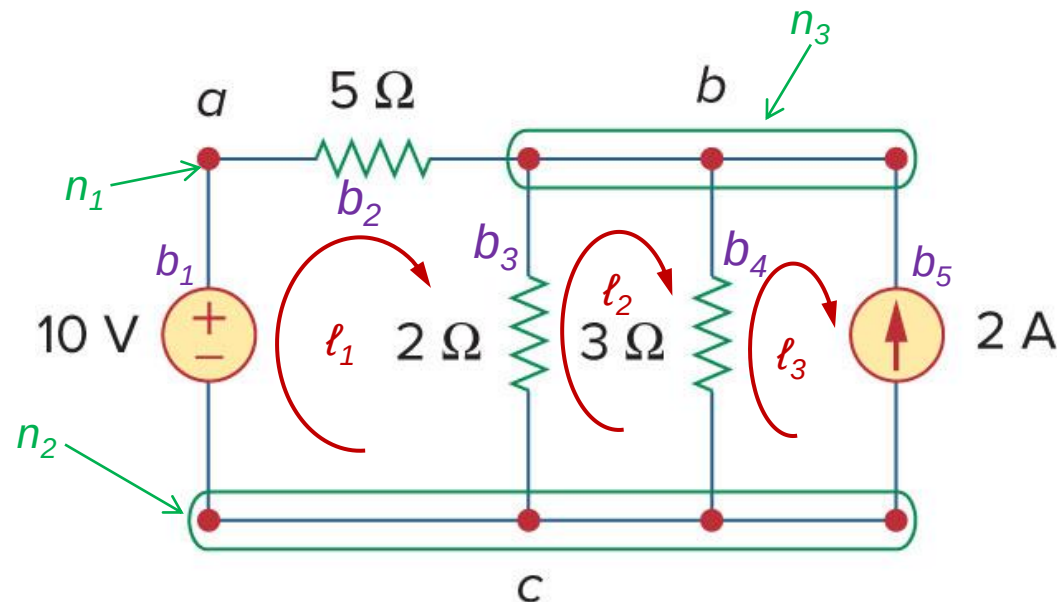
Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

Nodes, branches and loops/meshes

- Circuit elements can be interconnected in multiple ways.
- To understand this, we need to be familiar with some network topology concepts:
 - A **branch** represents a **single element** such as a voltage source or a resistor.
 - A **node** is the **point of connection** between two or more branches.
 - A **loop** is any **closed path** in a circuit.
 - A **mesh** is a loop that contains **no** other loop.

n_k = node k
 b_k = branch k
 ℓ_k = loop/mesh k

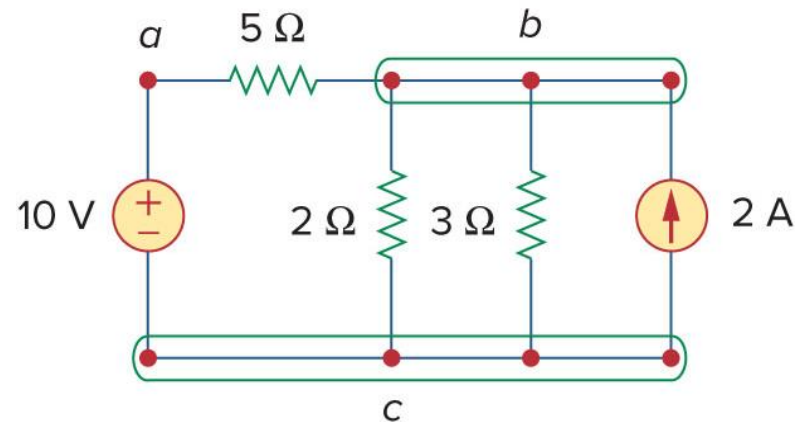
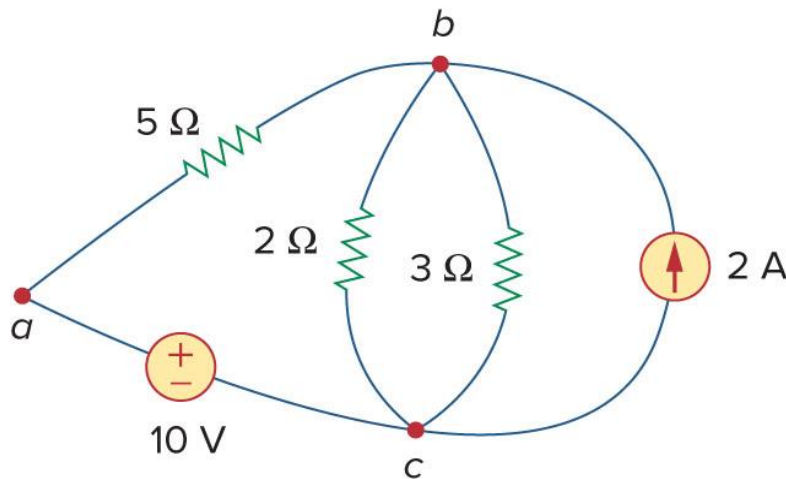


Series and parallel

- Two or more elements are in **series** if they **exclusively share a single node** and consequently carry the **same current**.
- Two or more elements are in **parallel** if they are connected to the **same two nodes** and consequently have the **same voltage** across them.

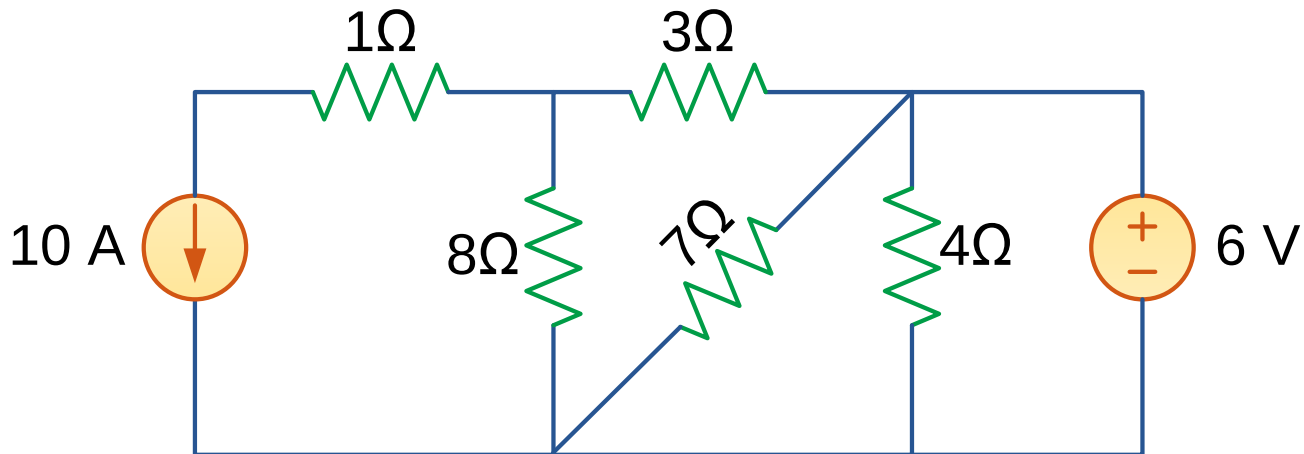
In the following example:

- Series: $5\ \Omega$ resistor and 10 V source.
- Parallel: 2 A source and $3\ \Omega$ and $2\ \Omega$ resistors.



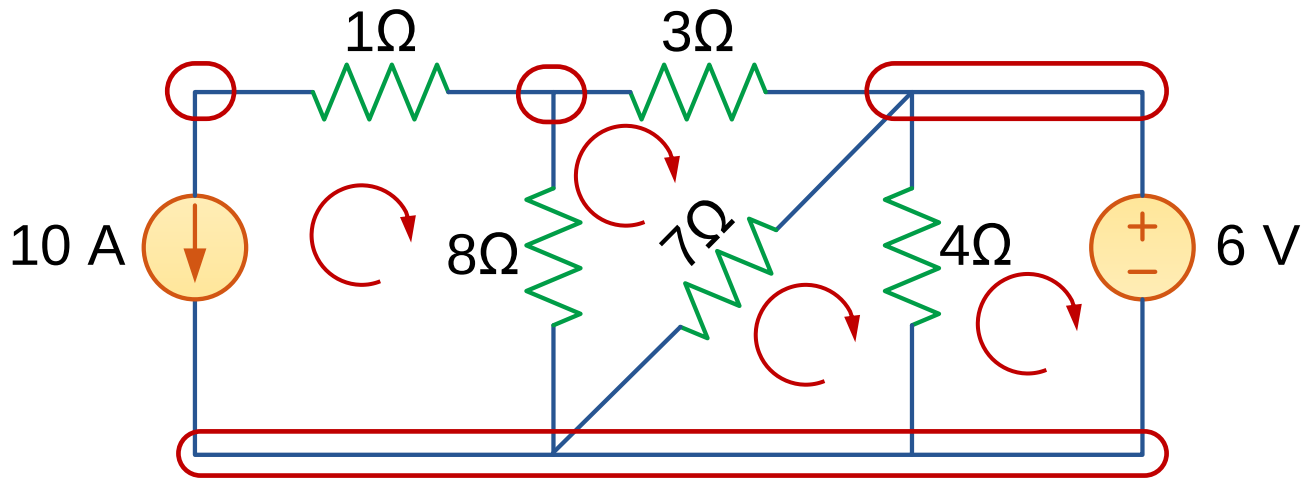
Exercise 1

- Count the number of branches, nodes, and meshes.
- Identify series and parallel elements.



Exercise 1

- Count the number of branches, nodes, and meshes.
- Identify series and parallel elements.



- 7 branches.
- 4 nodes.
- 4 meshes.
- Parallel: **6 V** source and **4 Ω** and **7 Ω** resistors.
- Series: **1 Ω** resistor and **10 A** source.

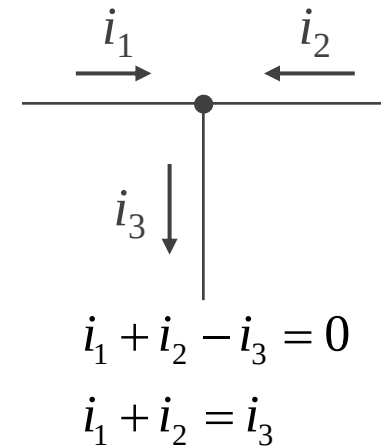
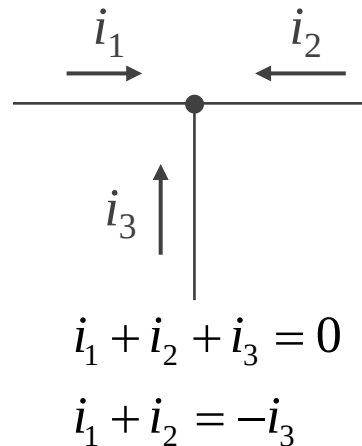
Kirchhoff's Current Law (KCL)

- The **algebraic sum of all currents entering a node is zero.**

$$\sum_{n=1}^N i_n = 0$$

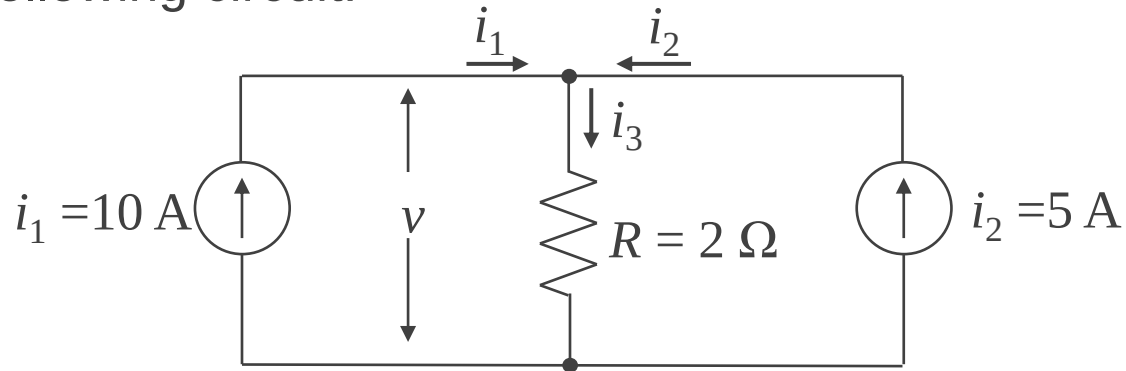
N is the number of branches connected to the node

- You may consider the currents **entering the node** to be **positive** and those **leaving the node** to be **negative** or vice versa.
- You may also consider that the **sum of currents entering the node is equal** to the **sum of currents leaving the node**.



Exercise 2

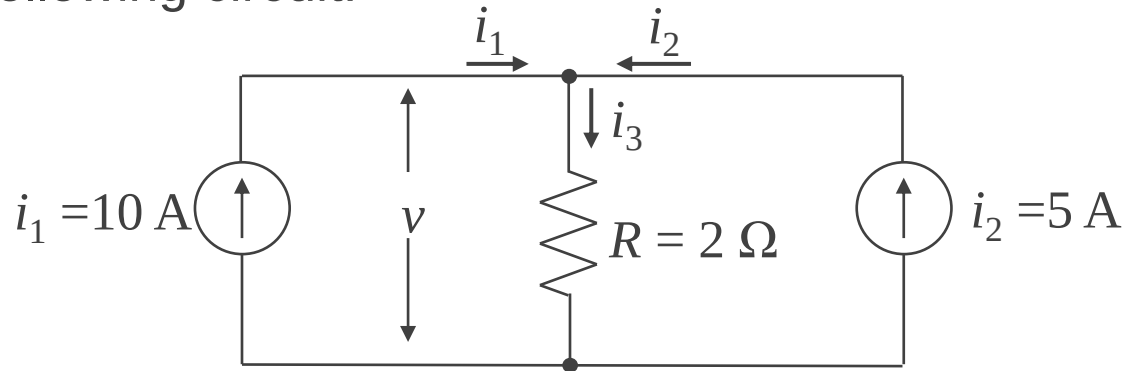
Given the following circuit:



- How much current flows through the resistor?
- What is the voltage across the resistor?

Exercise 2

Given the following circuit:



- How much current flows through the resistor?

$$\text{KCL: } i_1 + i_2 - i_3 = 0$$

$$i_3 = i_1 + i_2 = 10 \text{ A} + 5 \text{ A} = 15 \text{ A}$$

- What is the voltage across the resistor?

$$v = i_3 R = 15 \text{ A} \times 2 \Omega = 30 \text{ V}$$

Kirchhoff's Voltage Law (KVL)

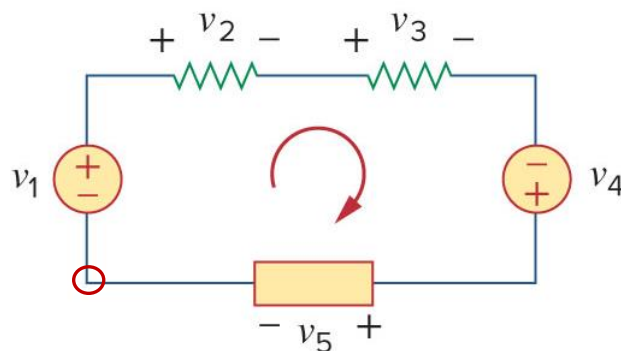
- The **algebraic sum of all voltage drops around a closed path** (or a loop) is **zero**.

$$\sum_{m=1}^M v_m = 0$$

M is the number of voltage drops in the loop

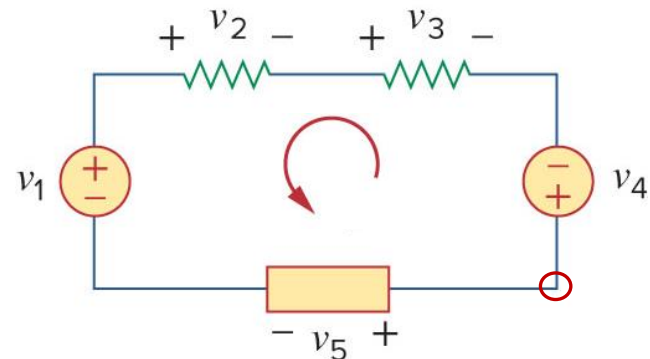
- You may **start from any node** in the loop and go around the loop **clockwise (CW)** or **counterclockwise (CCW)**.
- Use the **sign** of the terminal that you **first encounter** as you go around the loop.

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$



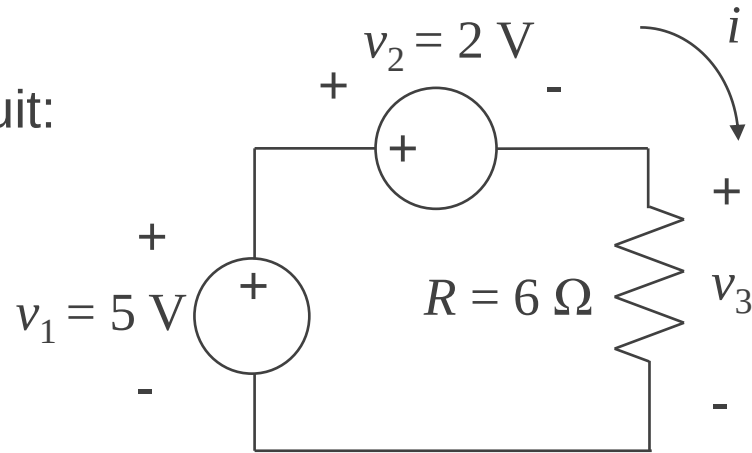
or

$$v_4 - v_3 - v_2 + v_1 - v_5 = 0$$



Exercise 3

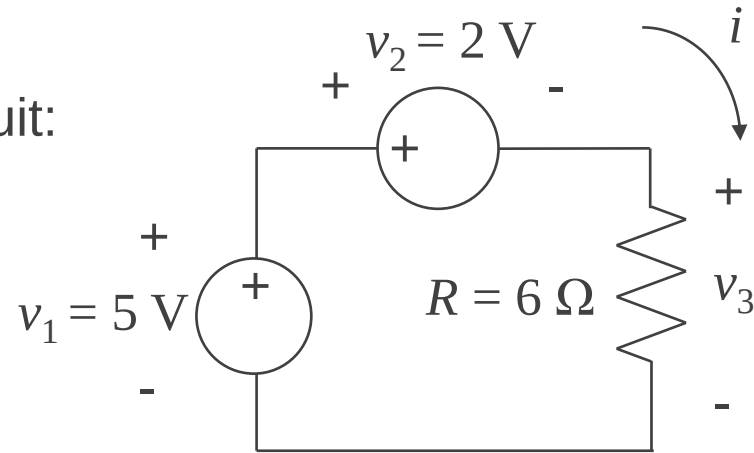
Given the following circuit:



- What is the voltage across the resistor?
- What is the current in the loop?

Exercise 3

Given the following circuit:



- What is the voltage across the resistor?

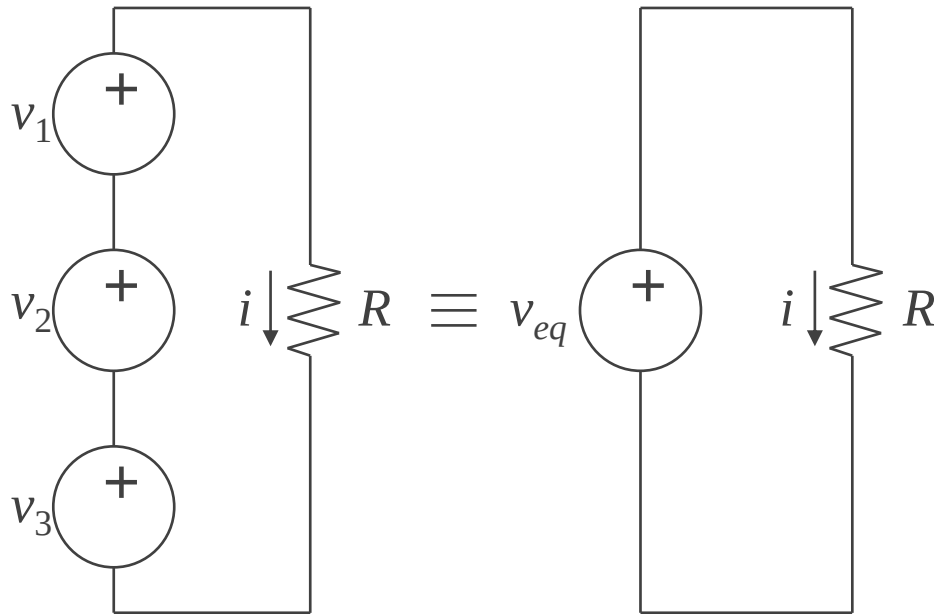
$$\text{KVL: } -v_1 + v_2 + v_3 = 0$$

$$v_3 = v_1 - v_2 = 5\text{ V} - 2\text{ V} = 3\text{ V}$$

- What is the current in the loop?

$$i = \frac{v_3}{R} = \frac{3\text{ V}}{6\ \Omega} = 0.5\text{ A}$$

Voltage sources in series



- Kirchoff's Voltage Law (KVL) gives:

$$-v_1 - v_2 - v_3 + iR = 0$$
$$(v_1 + v_2 + v_3) = iR$$

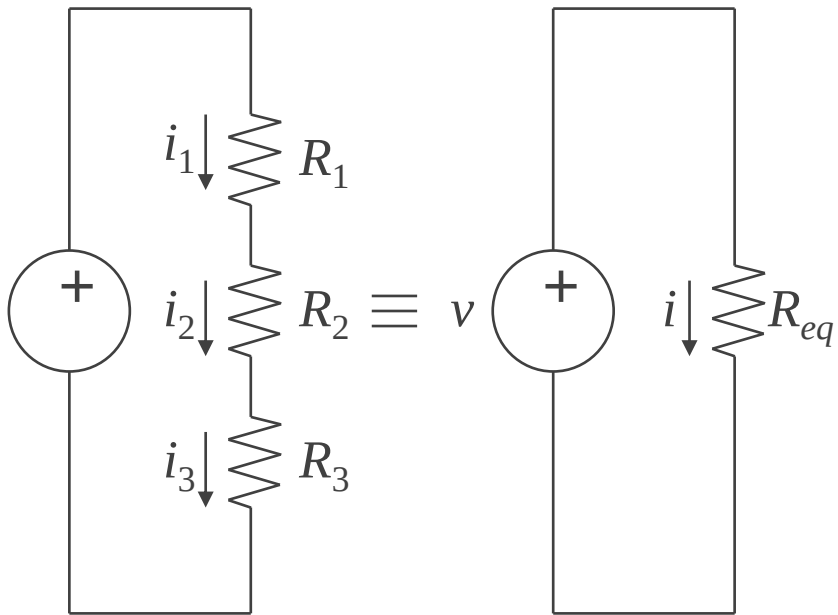
- Equivalent voltage:

$$v_{eq} = iR$$

$$v_{eq} = v_1 + v_2 + v_3$$

- Voltage sources in series are summed.

Resistors in series



- Current is the same in all resistors:

$$i_1 = i_2 = i_3 = i$$

- KVL gives:

$$-v + iR_1 + iR_2 + iR_3 = 0$$

$$v = i(R_1 + R_2 + R_3)$$

- Equivalent resistance:

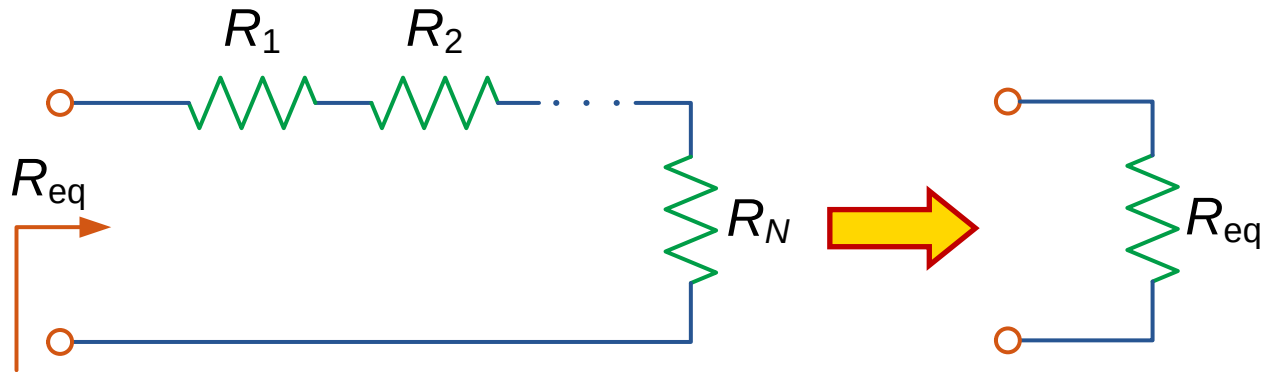
$$v = iR_{eq}$$

$$R_{eq} = R_1 + R_2 + R_3$$

- Resistors in series are summed.

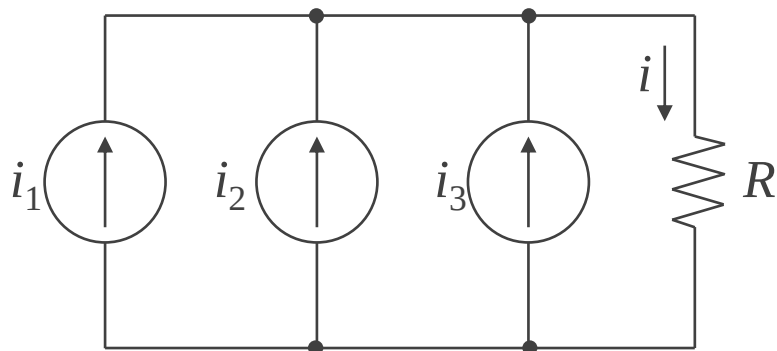
Resistors in series

- The **equivalent resistance** of any number of **series resistors** is the **sum** of the **individual resistances**.

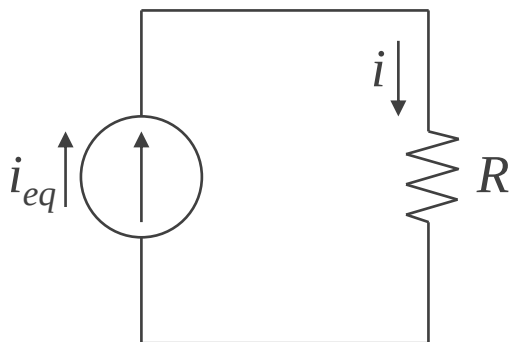


$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^N R_n$$

Current sources in parallel



|||



- We can find the current in the resistor by KCL:

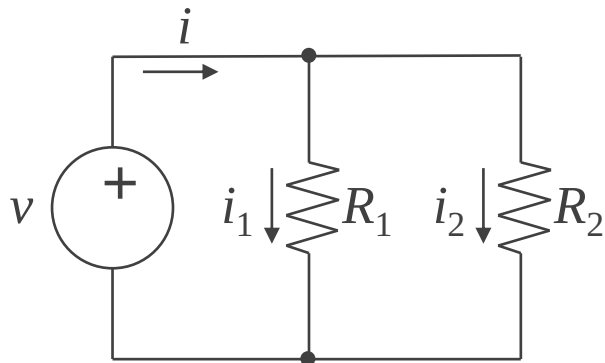
$$i_1 + i_2 + i_3 = i$$

- The equivalent current source would put the same current through the resistor, so:

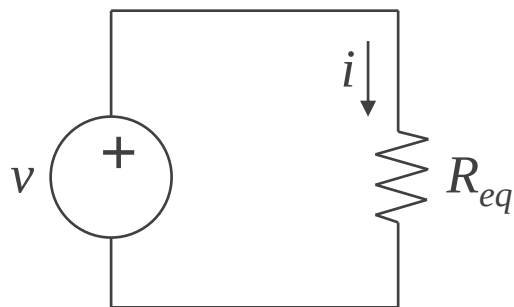
$$i_{eq} = i_1 + i_2 + i_3$$

- Current sources in parallel are summed.

Resistors in parallel



|||



- Voltages across the resistors are the same:

$$v = i_1 R_1 = i_2 R_2$$

Therefore:

$$i_1 = \frac{v}{R_1}; i_2 = \frac{v}{R_2}$$

- Using KCL:

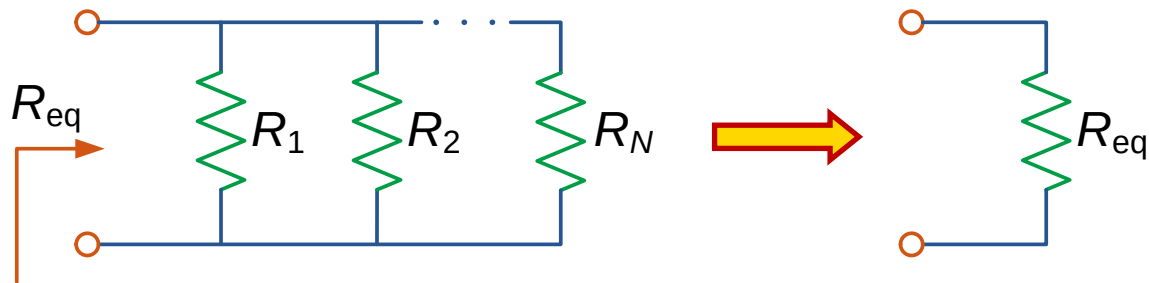
$$i = i_1 + i_2 = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\frac{i}{v} = \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$

Resistors in parallel

- The **reciprocal** of the **equivalent resistance** of any number of **parallel resistors** is the **sum** of the **individual reciprocal resistances**.



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

or

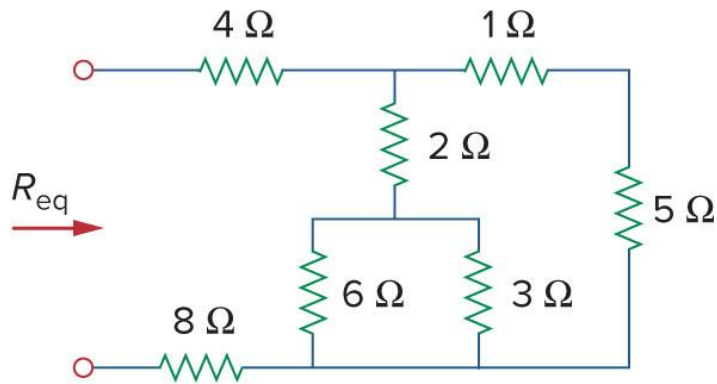
$$G_{eq} = G_1 + G_2 + \dots + G_N$$

or

$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}}$$

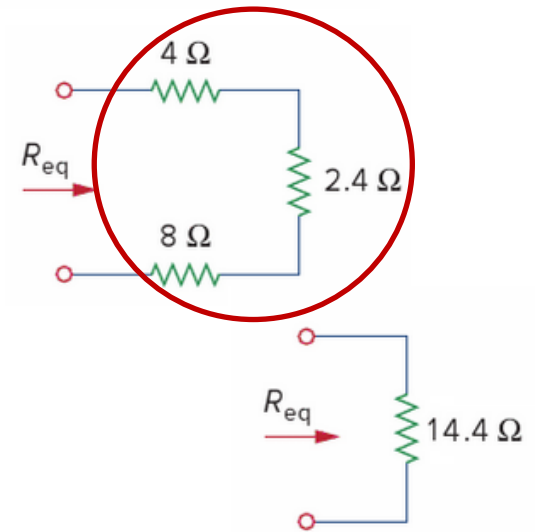
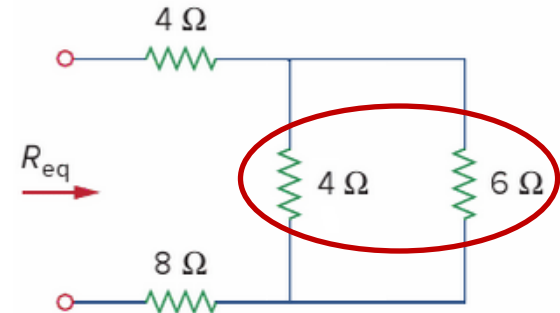
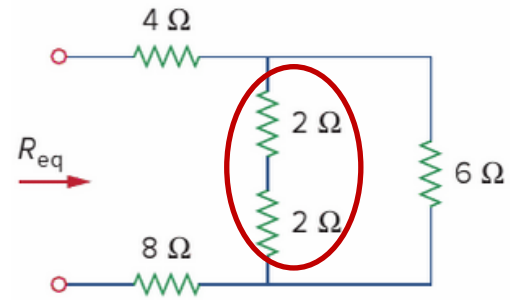
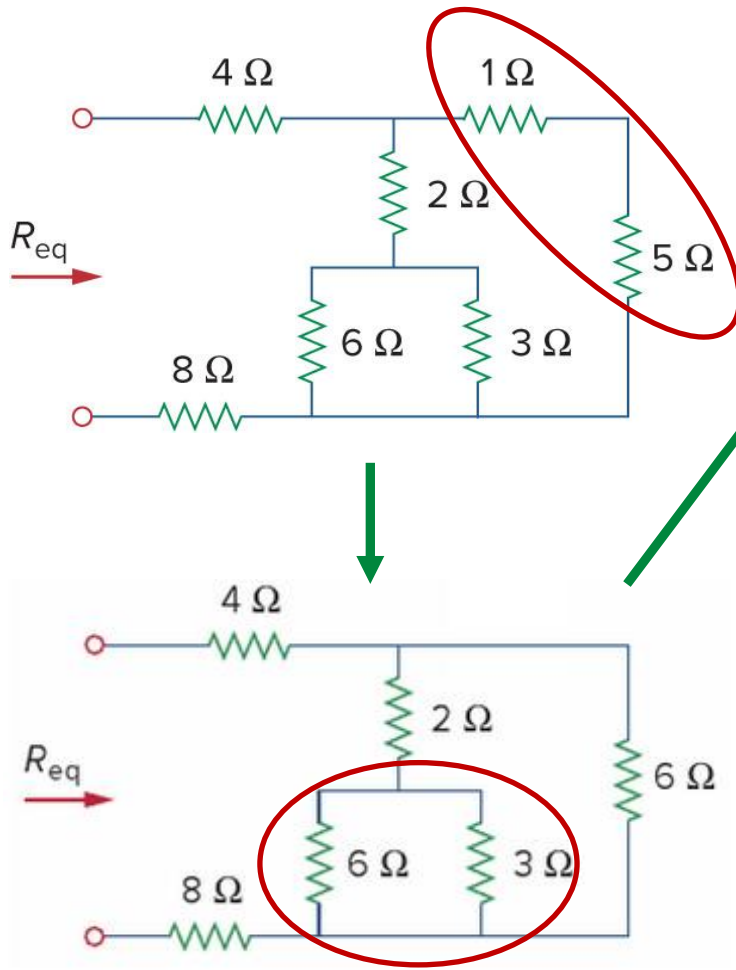
Exercise 4

Find the equivalent resistance R_{eq} .



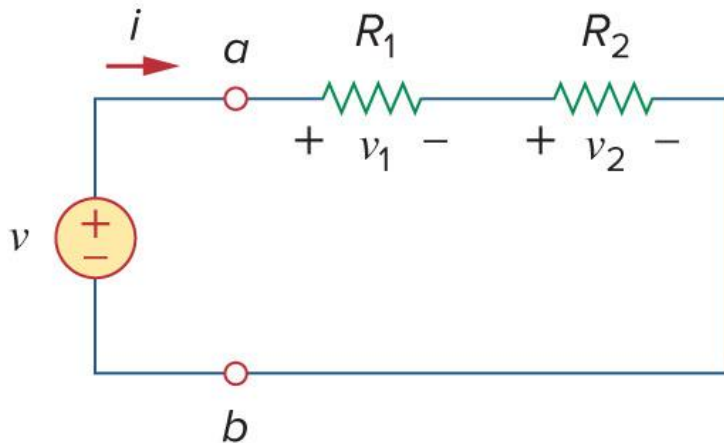
Exercise 4

Find the equivalent resistance R_{eq} .



Voltage divider

- A voltage divider is a simple circuit that divides a voltage **in proportion** to the **series resistances** (the higher the resistance, the higher the voltage).



- The current through the resistors is:

$$i = \frac{v}{R_1 + R_2}$$

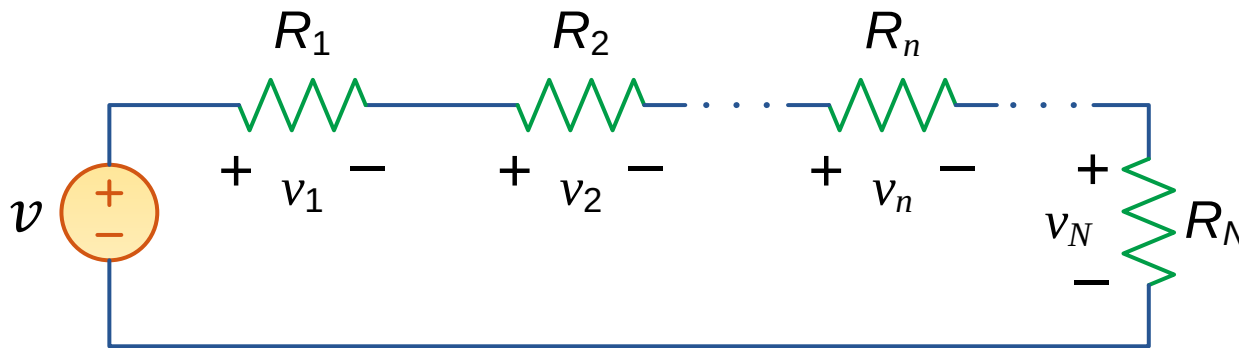
- The voltage across the resistors:

$$v_1 = iR_1; v_2 = iR_2$$

$$v_1 = v \frac{R_1}{R_1 + R_2}; v_2 = v \frac{R_2}{R_1 + R_2}$$

Voltage divider

- In general, the **voltage drop** across the n^{th} resistor in a **voltage divider** with N **series resistors** is obtained as follows:

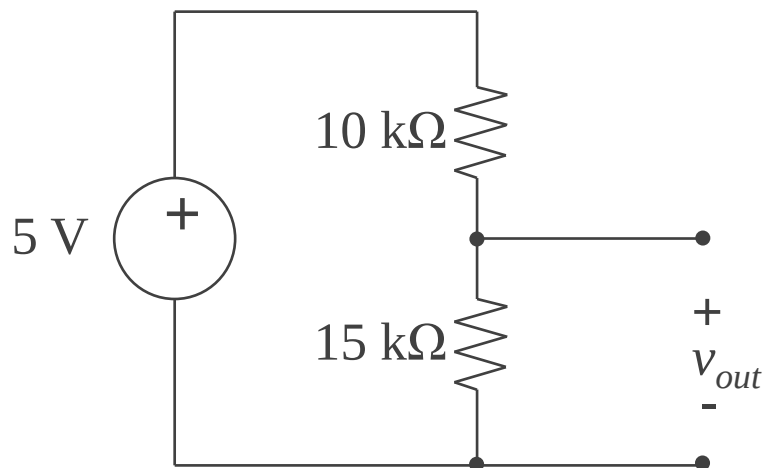


$$v_n = v \frac{R_n}{R_1 + R_2 + \dots + R_N} = v \frac{R_n}{R_{eq}}$$

Remark: As long as we know the input voltage to be divided, this voltage can be provided by any element or circuit, not necessarily a voltage source.

Exercise 5

- Find the voltage across the 15 k Ω resistor.

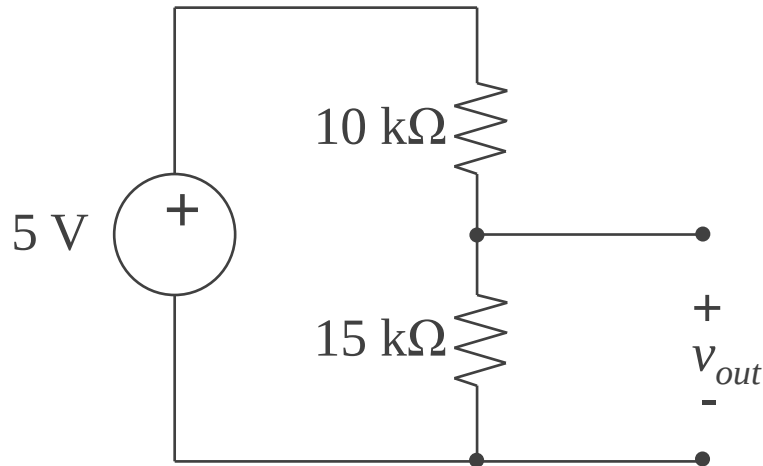


$$v_n = v \frac{R_n}{R_{eq}}$$

Exercise 5

- Find the voltage across the 15 kΩ resistor.

Using the voltage divider formula:



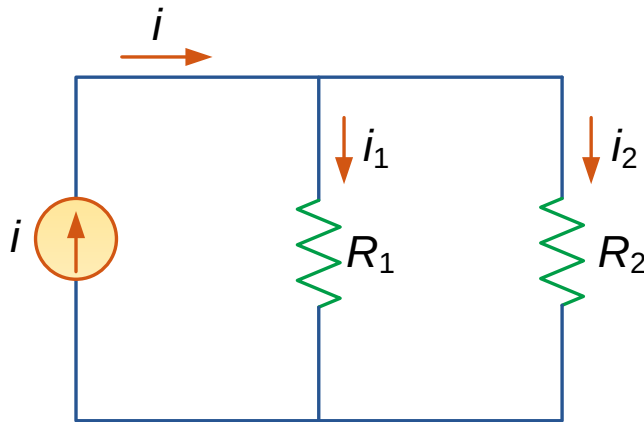
$$\begin{aligned} v_{out} &= v \frac{R_2}{R_1 + R_2} \\ &= 5 \times \frac{15,000}{10,000 + 15,000} \text{ V} \\ &= 5 \times \frac{15}{25} \text{ V} \\ &= 3 \text{ V} \end{aligned}$$



$$v_n = v \frac{R_n}{R_{eq}}$$

Current divider

- A current divider is a simple circuit that divides the current among **parallel resistors** in **inverse proportion** to their **resistance** (the higher the resistance, the lower the current).



- The voltage across the resistors is:

$$v = i(R_1 \parallel R_2)$$

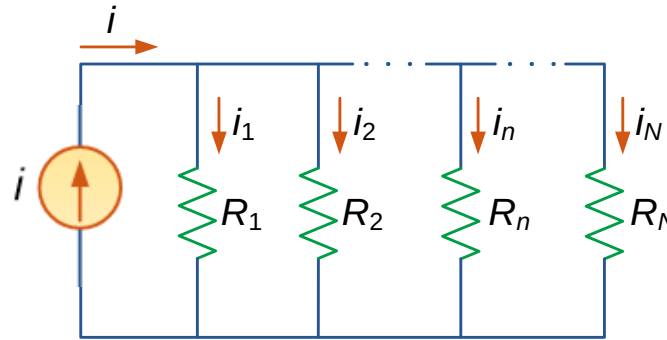
- The current through each resistor:

$$i_1 = \frac{v}{R_1} = i \frac{(R_1 \parallel R_2)}{R_1}$$

$$i_2 = \frac{v}{R_2} = i \frac{(R_1 \parallel R_2)}{R_2}$$

Current divider

- In general, the **current** through the n^{th} resistor in a **current divider** with N **parallel resistors** is obtained as follows:



$$i_n = i \frac{R_1 \parallel R_2 \parallel \dots \parallel R_N}{R_n} = i \frac{R_{eq}}{R_n}$$

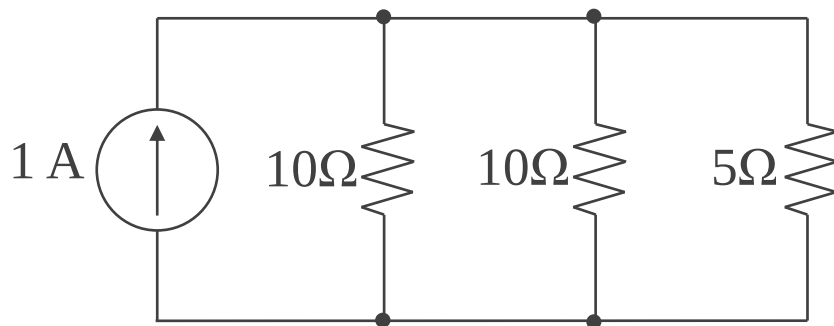
- If conductances are used instead of resistances, then:

$$i_n = i \frac{G_n}{G_1 + G_2 + \dots + G_N}, \text{ where } G_n = \frac{1}{R_n}$$

Remark: As long as we know the input current to be divided, this current can be provided by any element or circuit, not necessarily a current source.

Exercise 6

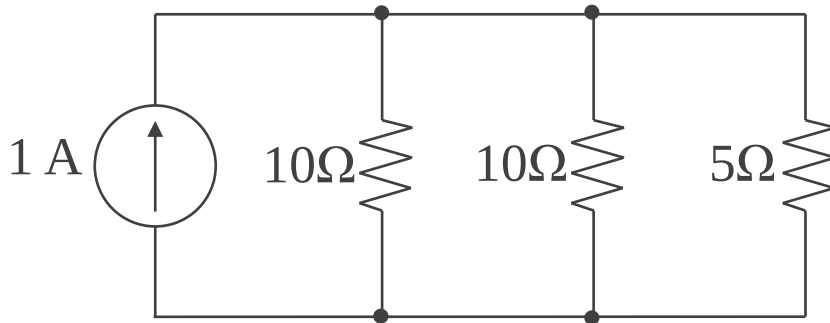
- Find the current in the 5 Ω resistor.



$$i_n = i \frac{R_{eq}}{R_n}$$

Exercise 6

- Find the current in the 5 Ω resistor.



- Find the parallel resistance:

$$R_{eq} = \frac{1}{\frac{1}{10} + \frac{1}{10} + \frac{1}{5}} = \frac{5}{2} = 2.5\Omega$$

- Current in 5 Ω resistor:

$$i_n = i \frac{R_{eq}}{R_n} = 1 \times \frac{2.5}{5} = 0.5A$$



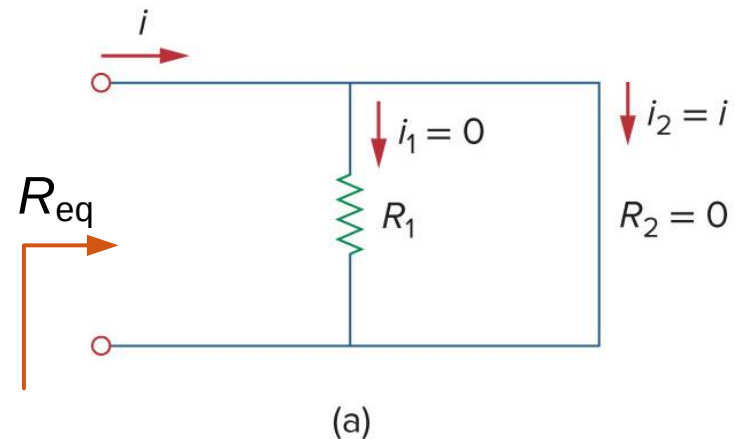
$$i_n = i \frac{R_{eq}}{R_n}$$

Current divider

Specific cases

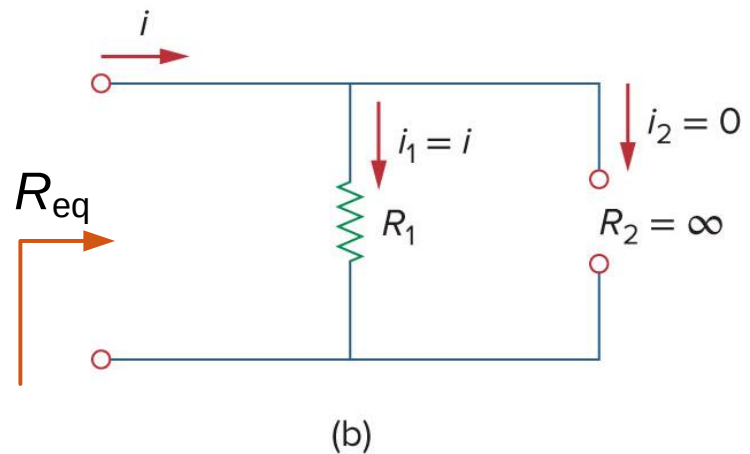
- **Short circuit:** The entire current i flows through the **smallest resistance** (short circuit), effectively bypassing R_1 .

$$R_{eq} = 0$$



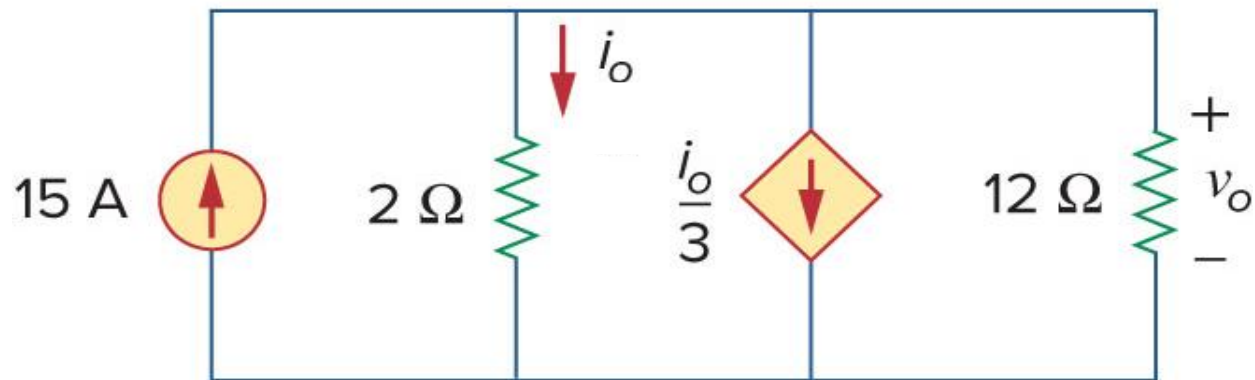
- **Open circuit:** The entire current i flows through the **smallest resistance** (R_1).

$$R_{eq} = R_1$$



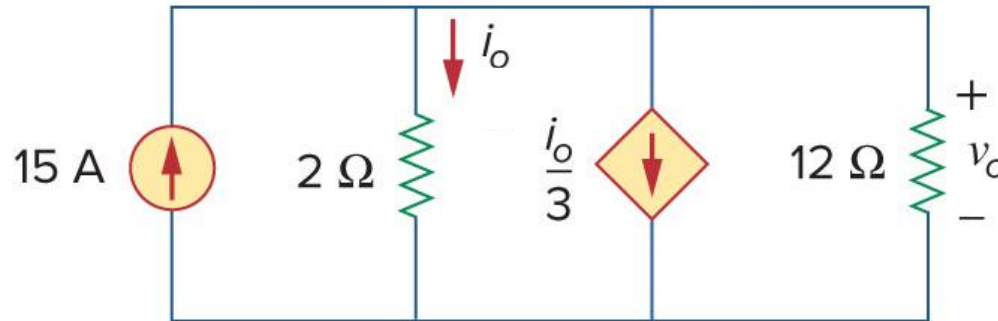
Exercise 7

Find v_o and i_o in the circuit below.



Exercise 7

Find v_o and i_o in the circuit below.



$$\text{KCL: } 15 \text{ A} = i_o + \frac{i_o}{3} + \frac{v_o}{12}$$

Since the 2Ω and 12Ω resistor are in parallel, they have the same voltage:

$$v_{2\Omega} = v_o = 2i_o$$

Substituting in the KCL equation:

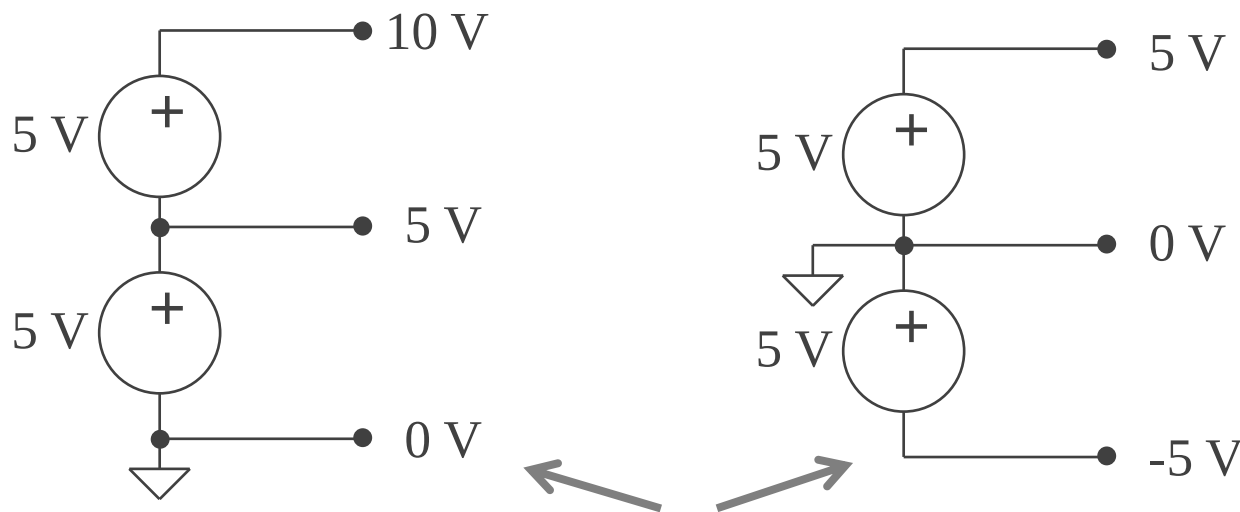
$$15 \text{ A} = i_o + \frac{i_o}{3} + \frac{2i_o}{12} = i_o \left(1 + \frac{1}{3} + \frac{1}{6} \right) = \frac{9}{6} i_o \rightarrow i_o = \frac{6 \times 15}{9} = 10 \text{ A}$$

$$v_o = 2i_o = 20 \text{ V}$$

Remark: There are many different ways to solve this (e.g. simplifying current sources, using current divider, etc.)

Circuit ground

- Voltage is a differential quantity, so we need a reference node, usually at zero volts.
- Any node in a circuit can be defined as zero volts.
- The **zero volt** point is referred to as the **circuit ground**.
- Symbol: 



Different choice of circuit ground leads to different voltages with respect to ground.

Earth

- An earthed ground is literally a connection to the earth, which provides an important role in electrical safety.
- Symbol: $\perp$



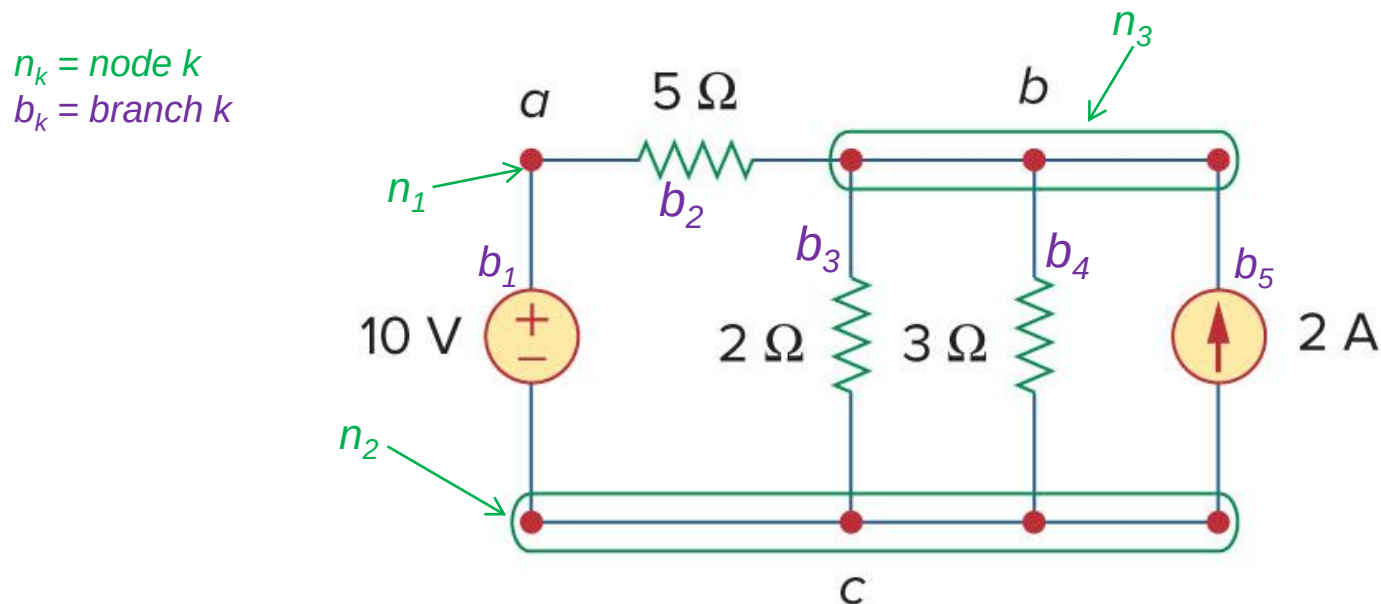
Circuit analysis

- Using Kirchhoff's laws and Ohm's law we can analyse any circuit to determine the currents and voltages; however, this might result in many simultaneous equations.
- The challenge of formal circuit analysis is to derive the **smallest set of simultaneous equations** that completely define the operating characteristics of a circuit (voltages and currents).
- Nodal and mesh analysis are two very useful methods for analysing any circuit.
 - They are based on the systematic application of Kirchhoff's laws.

Nodal analysis

Nodes

- A branch represents a single element such as a voltage source or a resistor.
- A node is the point of connection between two or more branches.



Nodal analysis

- It is based on KCL.
- We use **node voltages** (potential of each node) as the **main circuit variables**.
 - Voltages are relative to a reference node.
- Objective: To solve for these node voltages.
 - In general, an N -node circuit will need $N-1$ voltages and $N-1$ equations.
 - It will also require the solution of a $N-1$ system of equations.
 - KCL will be applied at each node except for one – the *reference node*.

Nodal analysis

Reference node

Any node can be chosen as the reference node.

Most common choices are:

- the ground node,
- top or bottom node, or
- node connected to the highest number of branches.

Nodal analysis

- Given a circuit with n nodes, the nodal analysis is accomplished via the following steps:

1. Select a node as the **reference node**.
2. Assign voltages $v_1, v_2, \dots, v_n$ to the remaining $n - 1$ nodes. These voltages are relative to the reference node.
3. Apply **KCL** to each of the $n - 1$ non-reference nodes.
 - For resistors, use Ohm's law to express the currents in terms of node voltages. Keep in mind the passive sign convention.
4. Solve the resulting $n - 1$ simultaneous equations to obtain the unknown node voltages.

NOTE: Always simplify the circuit before you start doing the analysis.

Nodal analysis

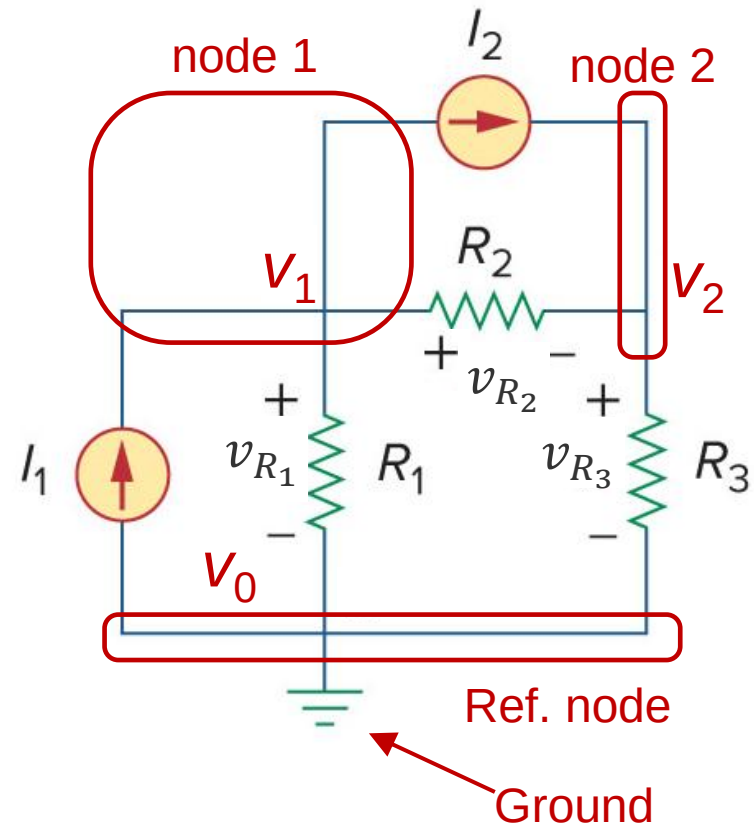
Before starting, **simplify** the circuit (if possible) and identify all the nodes.

1. Choose **ground** as **reference node** ($v_0 = 0$ V).
2. Assign voltages v_1 and v_2 to nodes 1 and 2.
 - Recall that:

$$v_{R_1} = v_1 - 0$$

$$v_{R_2} = v_1 - v_2$$

$$v_{R_3} = v_2 - 0$$



Nodal analysis

3. Apply KCL to nodes 1 and 2.

node 1: $I_1 = I_2 + i_1 + i_2$

node 2: $I_2 + i_2 = i_3$

- For resistors, use Ohm's law to express the branch currents in terms of node voltages. Keep in mind the passive sign convention.

$$i_n = \frac{v_{R_n}}{R_n} = \frac{v_{\text{higher}} - v_{\text{lower}}}{R_n}$$

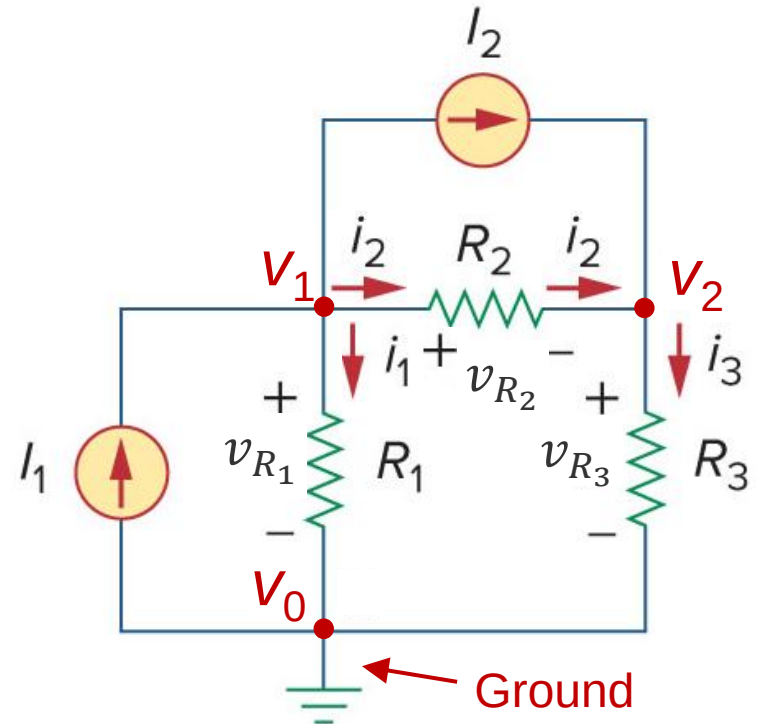
$$i_1 = \frac{v_{R_1}}{R_1} = \frac{v_1 - 0}{R_1}$$

$$i_2 = \frac{v_{R_2}}{R_2} = \frac{v_1 - v_2}{R_2}$$

$$i_3 = \frac{v_{R_3}}{R_3} = \frac{v_2 - 0}{R_3}$$

- Substitute back i_1 , i_2 , and i_3 into the node equations

4. Solve simultaneous equations (for v_1 and v_2).

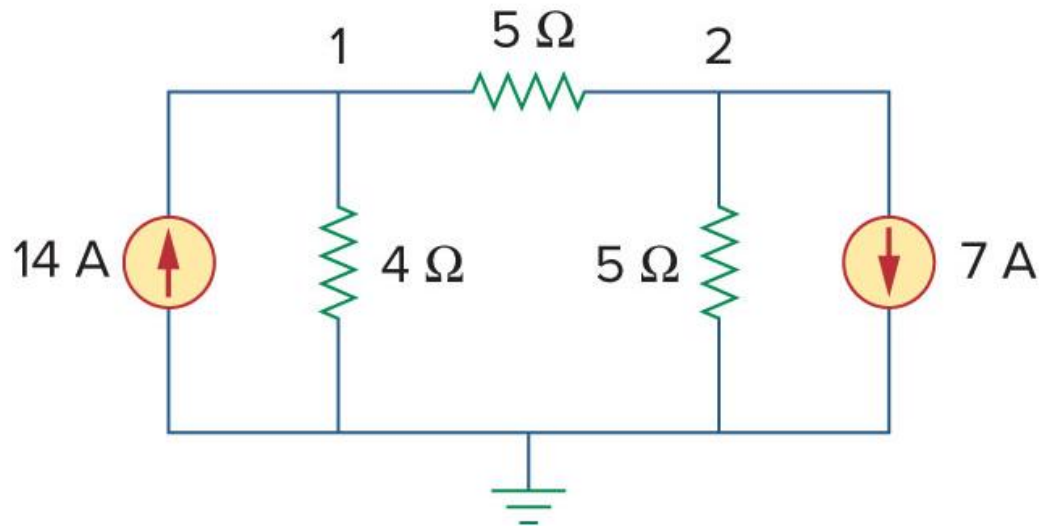


$$\begin{cases} I_1 = I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2} \\ I_2 + \frac{v_1 - v_2}{R_2} = \frac{v_2}{R_3} \end{cases}$$

Exercise 8

Obtain the node voltages in the circuit below and find the power dissipated in the 4Ω resistor.

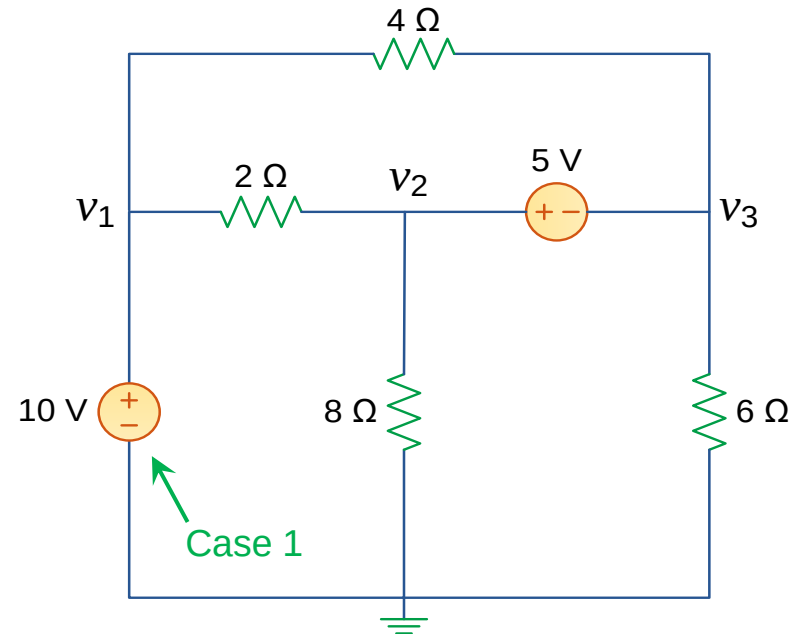
Solution in Exercises_solution_Topic_2.pdf



Nodal analysis with voltage sources

- Voltage sources generate or dissipate power at a specified voltage with whatever current is required.
 - The voltage is known at the terminals, but the current is not (and Ohm's law does not apply).
- There are 2 cases for nodal analysis with (independent or dependent) voltage sources.
 - **Case 1:** Voltage source is between the reference node and a non-reference node.
 - Set the voltage at the non-reference node to the voltage of the source.

$$v_1 = 10 \text{ V}$$

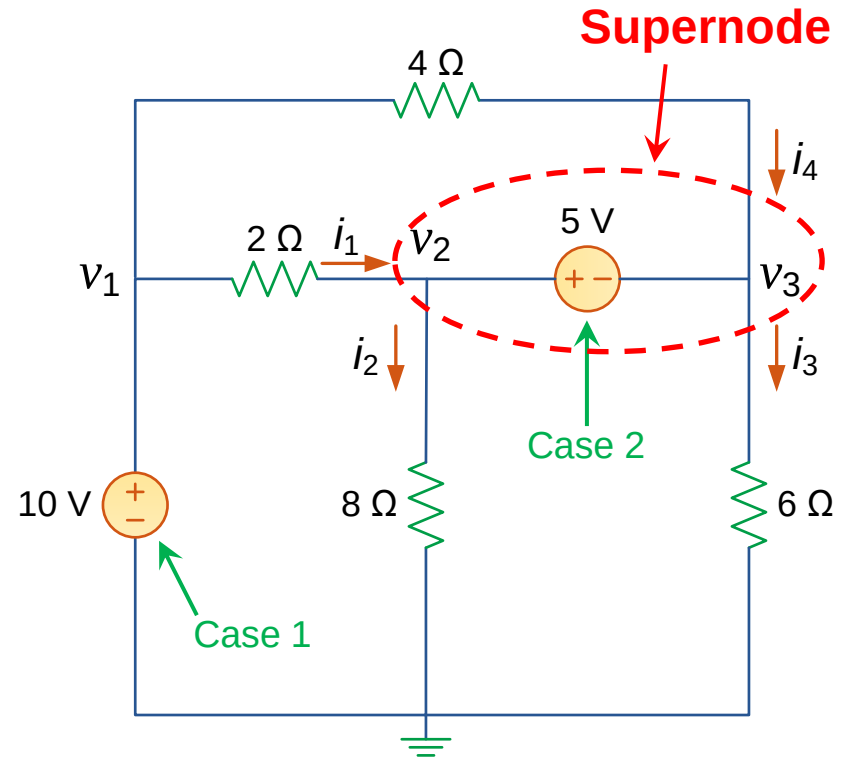


Nodal analysis with voltage sources

- **Case 2:** Voltage source is between two non-reference nodes.
 - The two nodes form a **supernode**.
 - The voltage across the voltage source can be expressed in terms of node voltages (KVL in bottom-right mesh).

$$v_2 - v_3 = 5 \text{ V}$$

A **supernode** is formed by **enclosing a voltage source** connected between **two non-reference** nodes and **any element** connected **in parallel** with it.



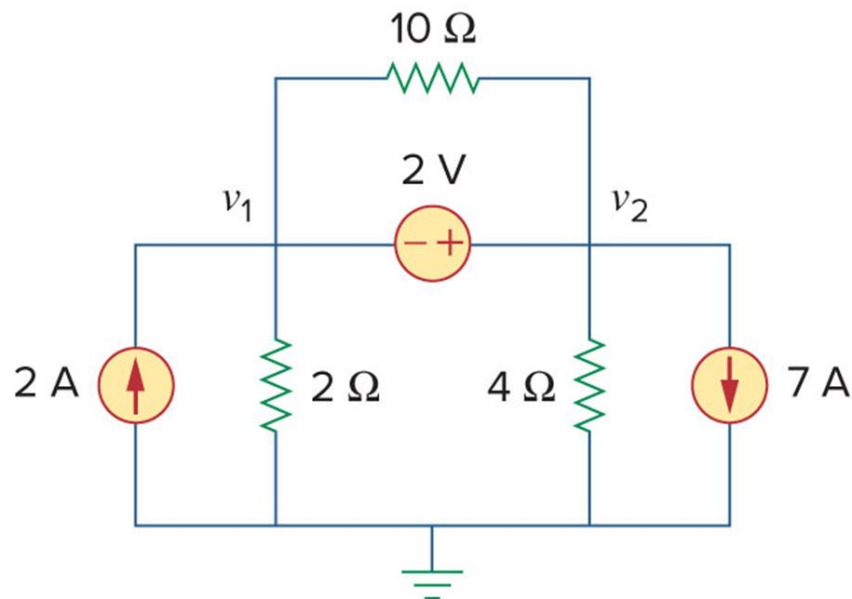
Supernode properties

1. The voltage source inside the supernode provides a **constraint equation** needed to solve node voltages.
2. A supernode has **no voltage** of its own.
3. A supernode requires the application of **both KCL** and **KVL**.

Exercise 9

Find the node voltages in the circuit below.

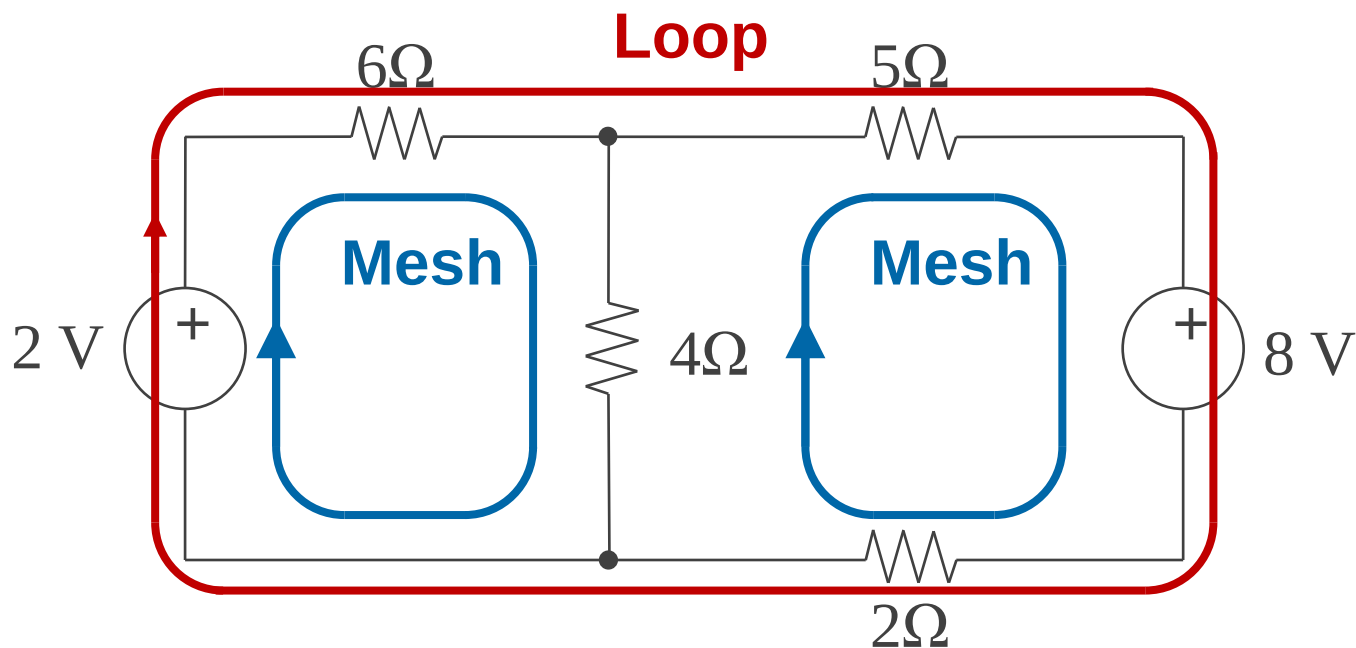
Solution in Exercises_solution_Topic_2.pdf



Mesh analysis

Meshes

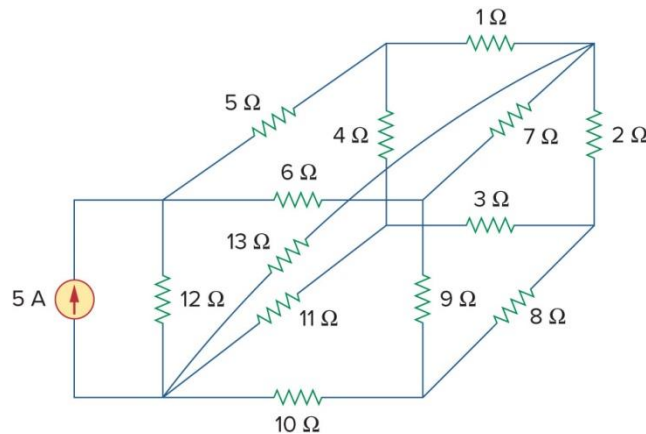
- A loop is any closed path in a circuit.
- A mesh is a loop that contains no other loop.



Mesh analysis

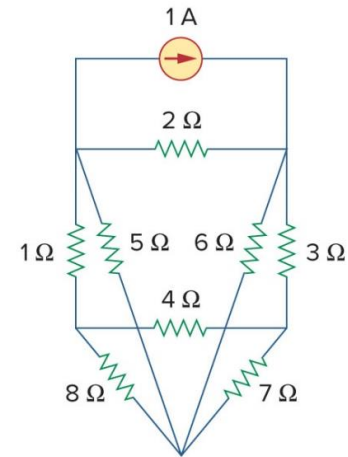
Planar vs non-planar circuits

- A planar circuit is a circuit that can be drawn in a plane with **no crossing branches**.
- Mesh analysis is only applicable to planar circuits.
 - Nodal analysis can be applied to both.

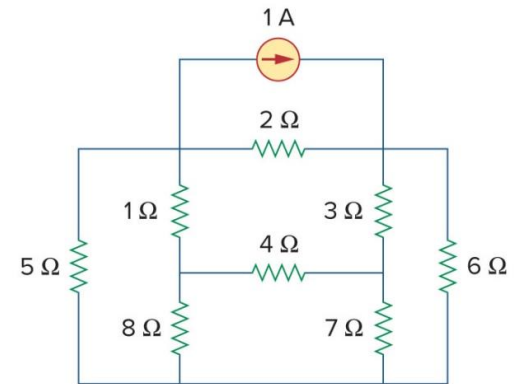


Non-planar circuit.

The branch with the 13Ω resistor prevents the circuit from being drawn without crossing branches.



(a)



(b)

Planar circuit.

(a) Original, (b) Redrawn circuit.

Mesh analysis

- It is based on KVL.
- We use **mesh currents** instead of element currents as the **main circuit variables**.
 - It reduces the number of equations that must be solved simultaneously.
- Objective: To solve for these mesh currents.
 - In general, an N-mesh circuit will need N currents and N equations.
 - It will also require the solution of a N system of equations.
 - KVL will be applied at each mesh.

Mesh analysis

- Given a circuit with n meshes, the mesh analysis is accomplished via the following steps:

1. Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes with a direction (generally clockwise).
2. Apply **KVL** to each of the n meshes (following the same direction as mesh currents).
 - For resistors, use Ohm's law to express the voltages in terms of mesh currents.
3. Solve the resulting n simultaneous equations to obtain the unknown mesh currents.

NOTE: Always simplify the circuit before you start doing the analysis.

Mesh analysis

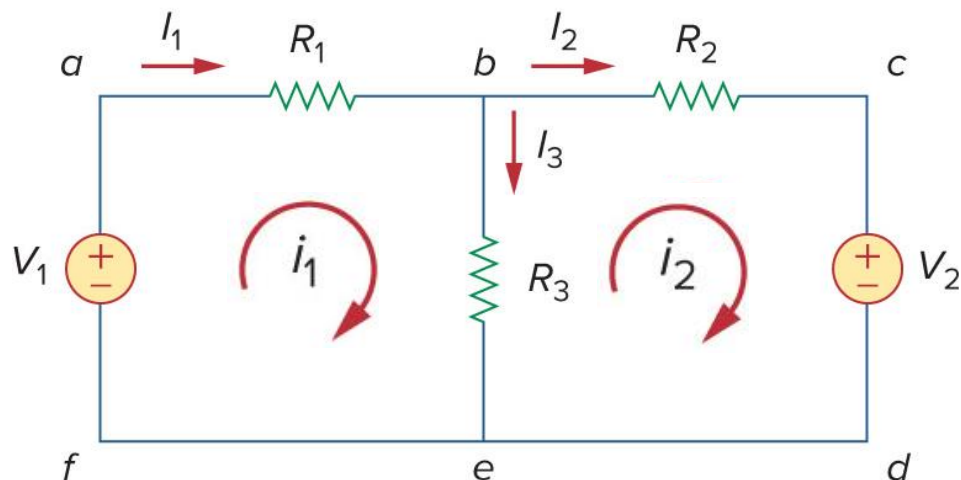
Before starting, **simplify** the circuit (if possible) and identify all the meshes.

1. Assign currents i_1 and i_2 to meshes 1 and 2.
 - Current for each branch is:

$$I_1 = i_1$$

$$I_2 = i_2$$

$$I_3 = i_1 - i_2$$



The current in the **common branch** between two meshes is the **difference** of the **two mesh currents** according to the current **direction** assigned to that branch.

NOTE: You can use I for **current branch** and i for **mesh current** or the other way round to distinguish between two types of currents, although most of the time we work directly with the mesh currents.

Mesh analysis

3. Apply KVL to meshes 1 and 2.

$$\text{mesh 1: } -V_1 + v_{R_1} + v_{R_3} = 0$$

$$\text{mesh 2: } -v_{R_3} + v_{R_2} + V_2 = 0$$

- For resistors, use Ohm's law to express the voltages in terms of mesh currents.

$$v_{R_1} = R_1 I_1 = R_1 i_1$$

$$v_{R_2} = R_2 I_2 = R_2 i_2$$

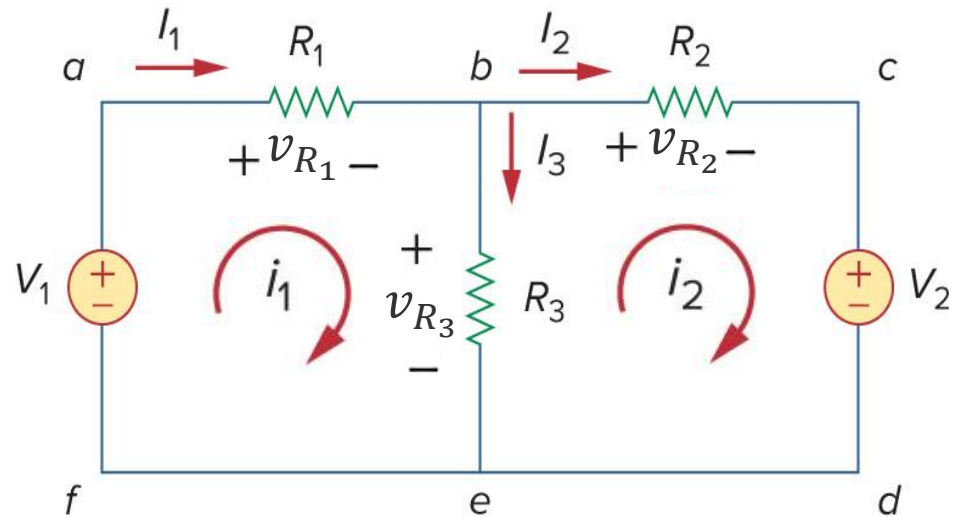
$$v_{R_3} = R_3 I_3 = R_3(i_1 - i_2)$$

- Substitute back v_{R_1} , v_{R_2} , and v_{R_3} into the mesh equations.



$$\begin{cases} -V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0 \\ -R_3(i_1 - i_2) + R_2 i_2 + V_2 = 0 \end{cases}$$

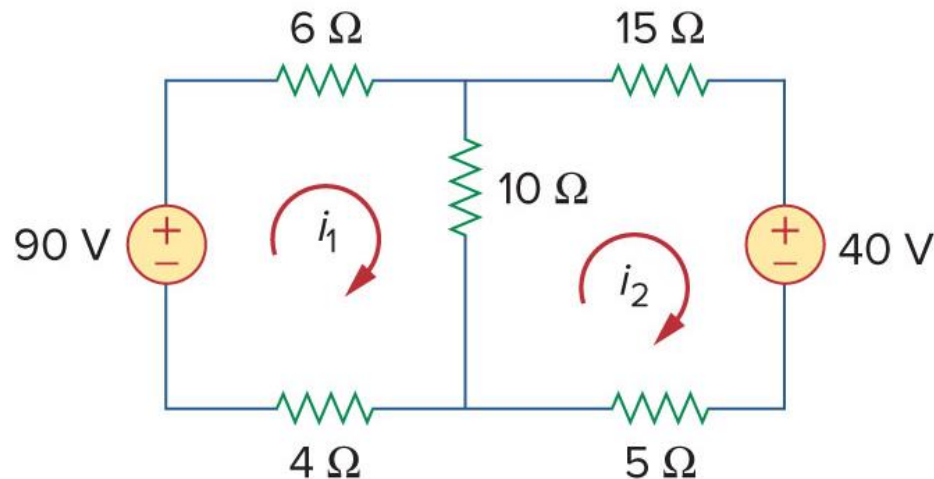
4. Solve simultaneous equations (for i_1 and i_2).



Exercise 10

Calculate the mesh currents i_1 and i_2 in the circuit below and find the power in the 40 V voltage source.

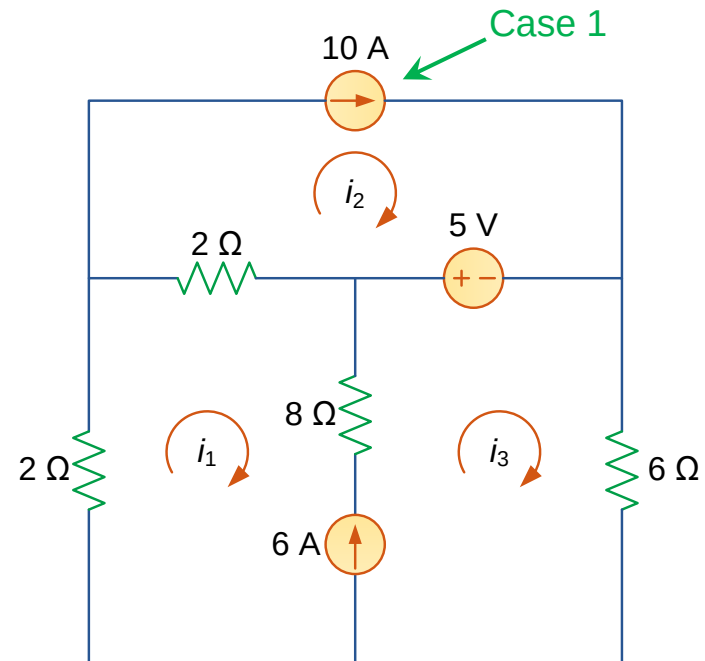
Solution in Exercises_solution_Topic_2.pdf



Mesh analysis with current sources

- Current sources generate or dissipate power at a specified current with whatever voltage is required.
 - The current is known at the terminals, but the voltage is not (and Ohm's law does not apply).
- There are 2 cases for mesh analysis with (independent or dependent) current sources.
- **Case 1:** Current source exists only in one mesh.
 - Set the mesh current to the current of the source.

$$i_2 = 10 \text{ A}$$

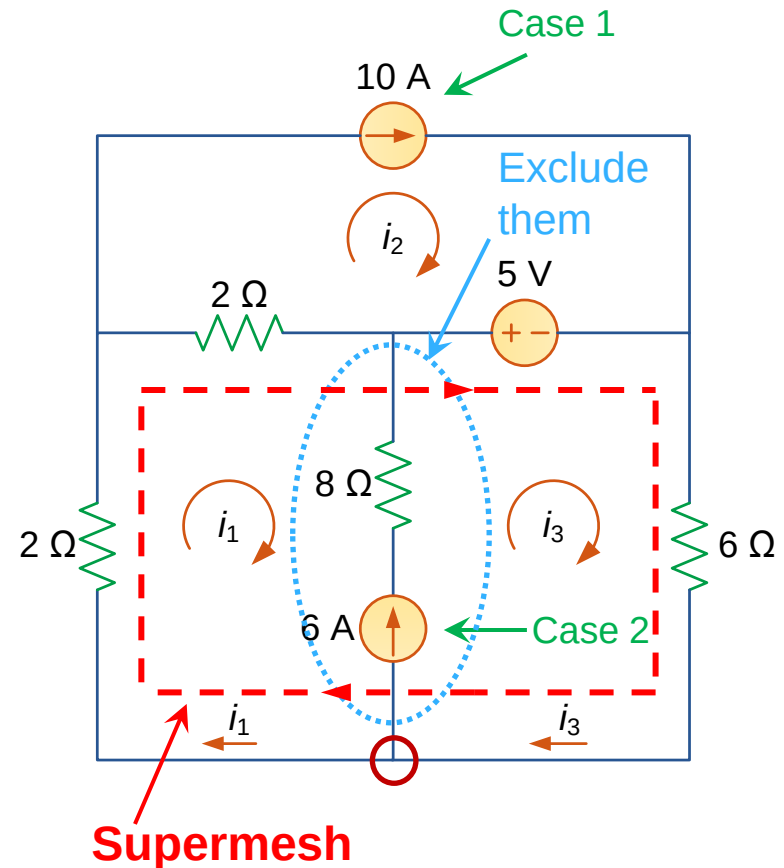


Mesh analysis with current sources

- **Case 2:** Current source is between two meshes.
 - The two meshes form a **supermesh** by excluding the branch with current source.
 - The current through the current source can be expressed in terms of mesh currents (KCL in bottom node).

$$i_3 = i_1 + 6 \text{ A}$$

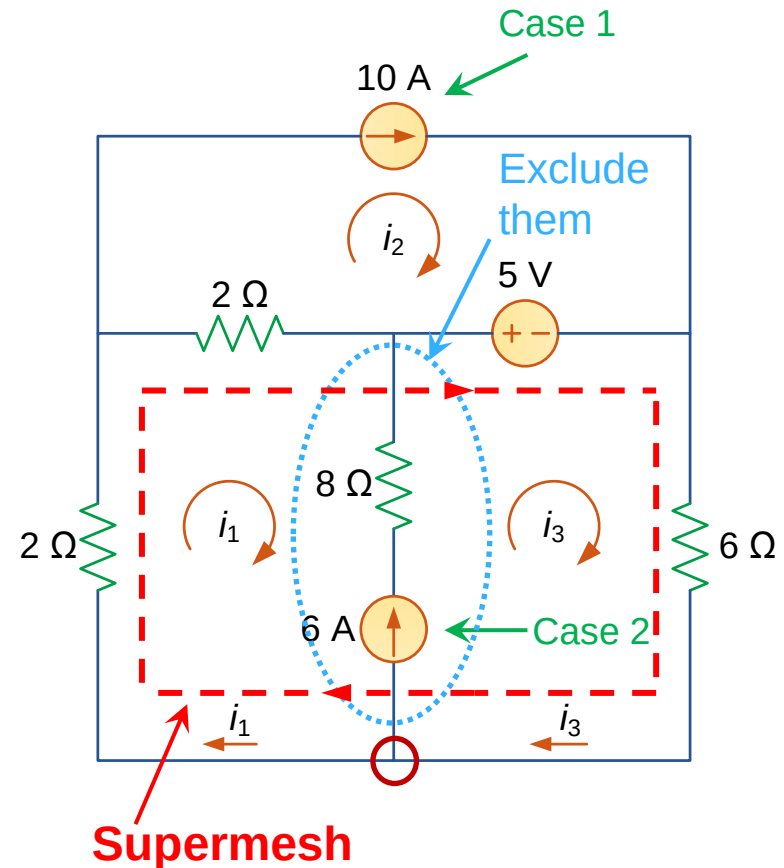
A **supermesh** is formed by **merging** two meshes and **excluding** the shared **current source** and **any element** connected **in series** with it.



Mesh analysis with current sources

Supermesh properties

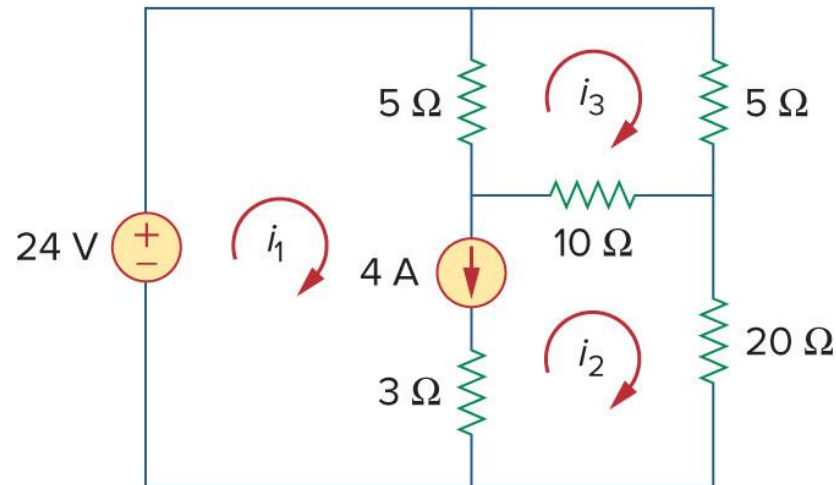
1. The current source inside the supermesh provides a **constraint equation** needed to solve mesh currents.
2. A supermesh has **no current** of its own.
3. A supermesh requires the application of **both KVL** and **KCL**.
4. **Intersecting** supermeshes in a circuit must be combined to form a **larger** supermesh.



Exercise 11

In the circuit below, use mesh analysis to determine i_1 , i_2 , and i_3 .

Solution in Exercises_solution_Topic_2.pdf



Nodal analysis vs mesh analysis

- We aim to select the method that results in the smaller number of equations.
 - We prefer **nodal analysis** for circuits with **fewer nodes** than meshes.
 - We prefer **mesh analysis** for circuits with **fewer meshes** than nodes.

$$\begin{aligned}
 & (x-2)^2(y-2x+2)^2(y+2x-10)^2(x-4)^2(y-2x+8)^2(y+2x-16)^2\left(y-3-3\left\lfloor\frac{x-11}{2}\right\rfloor\right)^2(x-8)^2 \\
 & \cdot\left(y-2-3\left\lfloor\frac{x-8}{2}\right\rfloor\right)^2(x-11)^2\left(y-\frac{1}{2}x+\frac{5}{2}-3\left\lfloor\frac{x-11}{2}\right\rfloor\right)^2\left(y+\frac{1}{2}x-\frac{17}{2}-3\left\lfloor\frac{x-11}{2}\right\rfloor\right)^2(x-15)^2 \\
 & \cdot\left(y-4-3\left\lfloor\frac{x-14}{2}\right\rfloor\right)^2(y-2x+52)^2(x-17)^2(y+x-21)^2(x-19)^2(y-x+17-3\lfloor x-20\rfloor)^2 \\
 & \cdot\left(y+x-23-3\lfloor x-20\rfloor\right)^2(y-x+19-3\lfloor x-21\rfloor)^2(y-3-3\lfloor x-21\rfloor)^2(x-25)^2\left(y+\frac{1}{4}x-\frac{41}{4}-3\left\lfloor\frac{x-25}{2}\right\rfloor\right)^2 \\
 & \cdot\left(y-\frac{1}{8}x-\frac{1}{8}-3\left\lfloor\frac{x-25}{2}\right\rfloor\right)^2\left(y+\frac{5}{8}x-\frac{151}{8}-3\left\lfloor\frac{x-25}{2}\right\rfloor\right)^2(y-2x+54)^2(y+2x-62)^2\left(y-3-3\left\lfloor\frac{x-57}{2}\right\rfloor\right)^2 \\
 & \cdot(x-31)^2(y+x-35)^2(x-33)^2(x-34)^2\left(y+\frac{1}{2}x-21-3\left\lfloor\frac{x-34}{2}\right\rfloor\right)^2\left(y-\frac{1}{2}x+15-3\left\lfloor\frac{x-34}{2}\right\rfloor\right)^2 \\
 & \cdot((x-38)^2+(y-3)^2-1)^2(x-40)^2(y+2x-84)^2(y-2x+80)^2(x-42)^2(x-43)^2\left(y-2-3\left\lfloor\frac{x-43}{2}\right\rfloor\right)^2 \\
 & \cdot(y-3-|x-47|)^2((x-47)^2+(y-3+\sqrt{y^2-6y+9})^2+(y^2-6y+8+\sqrt{y^4-12y^3+52y^2-96y+64})^2=0
 \end{aligned}$$

Questions?

